

# Project brief: Extending Theorem 2 (Robust Trust) beyond finite $M$ and $\Theta$

## Goal

Prove the **existence** part of Theorem 2 in *Robust Trust* (Dworczak–Smolin) **without assuming** that the adviser-belief support ( $M$ ) and the agent type space ( $\Theta$ ) are finite.

Equivalently: establish that the agent's robust max–min problem admits a **robustly rationalizable** optimal strategy (a saddle point / zero-sum equilibrium with the misaligned adviser) when ( $M$ ) and ( $\Theta$ ) may be infinite, within the paper's measurable, compact-metric setup.

## Context

- The paper models an agent choosing actions under uncertainty, combining private information ( $\theta$ ) with advice from an informed adviser.
- The adviser is truthful with probability ( $\alpha$ ) and adversarial (arbitrary messaging) with probability ( $1-\alpha$ ).
- Theorem 2 has two parts:
- **Optimality:** any robustly rationalizable strategy is optimal.
- **Existence (as stated):** a robustly rationalizable strategy exists if ( **$M$** ) and ( **$\Theta$** ) are finite.
- The finiteness assumption is used to make a direct minimax argument (via Sion-type conditions) technically straightforward. The project is to remove that crutch.

## What we need

1. **A precise generalized statement** of Theorem 2 (quantifiers + assumptions), keeping the paper's primitives intact:
  - ( $\Omega$ ) finite; ( $A$ ) compact metric; ( $\Theta$ ) compact metric; bounded utility ( $u$ ) (continuous in ( $a$ )); measurability throughout.
  - Only ( $M$ ) and ( $\Theta$ ) are allowed to be infinite.
2. **Strategy-space formalization for infinite settings**
  - Model ( $\sigma$ ) and ( $\beta$ ) as measurable stochastic kernels.
  - Choose topologies (typically weak/weak\* topologies on probability measures, product topologies on kernel spaces).
3. **Existence machinery**
  - Show convexity and compactness (or sequential compactness/tightness) of the relevant strategy spaces under the chosen topology.
  - Verify continuity or suitable semi-continuity of the payoff functional in each argument.
  - Apply an appropriate minimax / saddle-point theorem (Sion, Fan, Glicksberg-type, or an infinite-dimensional extension), or construct a saddle point directly.

#### 4. **No assumption smuggling**

- Any additional technical conditions needed (for example, continuity in  $(\theta)$ , continuity of type kernels  $(f(\cdot|\omega))$ , existence of measurable maximizers, compactness of certain induced correspondences) must be explicitly flagged as **Needed assumption**, not silently adopted.

#### 5. **Fallback: counterexample if needed**

- If a robustly rationalizable strategy can fail to exist under the baseline assumptions with infinite  $(M)$  and/or  $(\Theta)$ , produce a minimal counterexample.
- Then propose the weakest additional condition(s) under which existence is restored.

## Deliverables

### 1. **Generalized Theorem 2 statement** (optimality + existence) with explicit assumptions and quantifiers.

### 2. **Proof package**

- A clean proof of the **optimality direction** that does not rely on finiteness.
- A fully rigorous proof (or a clearly identified counterexample) for the **existence direction** when  $(M)$ ,  $(\Theta)$  are infinite.

### 3. **Technical appendix of lemmas** (as needed)

- Compactness/tightness results for the strategy spaces.
- Continuity or semi-continuity of payoff as a function of kernels.
- A precise minimax/saddle-point theorem statement with conditions checked line-by-line.

### 4. **Assumption ledger**

- A short list separating:
  - assumptions already present in the paper's model section, and
  - additional assumptions that turn out to be necessary for the infinite-space existence proof.

### 5. **Certification view (optional but valuable)**

- If existence holds, an explicit "certificate" template: how to exhibit  $(\beta^*)$  that makes a proposed  $(\sigma^*)$  robustly rationalizable.